Data Engineering and Feature Extraction for Real-Estate Valuation

Paris and Ile-de-France

Technical Deliverable

Acquisition, Preparation, Entity Resolution, Geospatial Enrichment, Feature Engineering, Quality Control and Modeling-Ready Data Products

Scope. This document defines the end-to-end data strategy for a machine-learning platform that estimates residential sale prices and rents in Paris and Ile-de-France. It covers nine core data families: DVF, BAN, Cadastre/BDNB, DPE, INSEE, Ile-de-France Mobilites, OpenStreetMap/APUR, rental observations, and macroeconomic variables.

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Prepared as a project implementation and architecture deliverable

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1 Executive Summary

The objective is to create a trustworthy, reproducible and extensible data foundation for an automated valuation model (AVM) covering residential properties in Paris and the wider Ile-de-France region. The central principle is that no single source describes a property sufficiently well. Instead, a property must be reconstructed by combining transaction, address, parcel, building, energy, neighborhood, mobility, amenity, rental and macroeconomic information.

The core modeling equation is:

$$\widehat{P}_{i,t} = f\left(X_i^{property}, X_i^{building}, X_{i,t}^{location}, X_t^{market}\right), \tag{1}$$

where $P_{i,t}$ is the transaction price or rent for property i at time t.

A production-quality system must do more than download files. It must guarantee:

- reproducibility of every data snapshot;
- stable internal identifiers across heterogeneous sources;
- explicit quality and match-confidence indicators;
- temporal correctness, so no future information leaks into historical training rows;
- spatial correctness, including appropriate coordinate reference systems and spatial joins;
- legal traceability, including source, licence and permitted use;
- model-ready features that can be recomputed consistently for both training and production inference.

Core design rule

Raw source records are never used directly as model inputs. Every source passes through a **Bronze** → **Silver** → **Gold** pipeline. Bronze preserves source fidelity; Silver standardizes and resolves entities; Gold contains leakage-safe analytical features and targets.

2 Business and Modeling Objectives

2.1 Primary prediction tasks

The platform should support at least four related but distinct prediction tasks:

$$\widehat{P}_{i,t}^{sale} = f_{sale}(X_{i,t}), \tag{2}$$

$$\widehat{P}_{i,t}^{rent} = f_{rent}(X_{i,t}), \tag{3}$$

$$\widehat{p}_{i,t}^{sale/m^2} = g_{sale}(X_{i,t}), \tag{4}$$

$$\widehat{\Delta P}_{z,t+h} = h(X_{z,t}, h), \tag{5}$$

where z denotes a spatial market zone and h a forecasting horizon.

Sale valuation, rental valuation, and market forecasting should be treated as separate tasks because they have different targets, sampling mechanisms, reporting delays and biases.

2.2 Model-ready analytical unit

For a simple residential sale model, one Gold row should correspond to one economically interpretable property transaction. For rental modeling, one Gold row should correspond to one rental observation with a clearly defined target such as asking rent excluding charges or signed rent excluding charges.

2.3 Target output

A production response should include more than a point estimate:

- estimated total price;
- estimated price per square meter;
- lower and upper prediction bounds;
- confidence score;
- number and quality of local comparables;
- major explanatory factors;
- data freshness indicators.

3 End-to-End Data Architecture

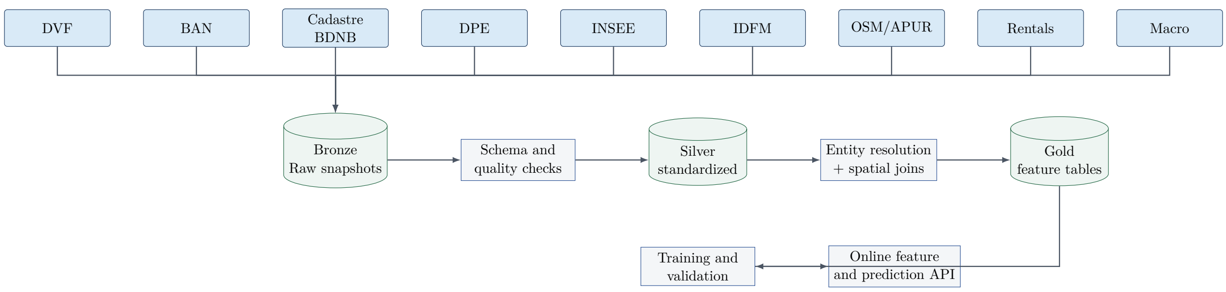

Figure 1: Logical data architecture for the valuation platform.

3.1 Bronze, Silver and Gold contracts

Layer	Purpose	Allowed transformations	Typical outputs
Bronze	Preserve exact source snapshots and provenance	Compression, checksum, metadata sidecar only	Original CSV, GeoJSON, GPKG, ZIP, API JSON
Silver	Standardize, clean and resolve identifiers	Type casting, normalization, deduplication, geocoding, CRS conversion	Transactions, addresses, parcels, buildings, DPE, POIs
Gold	Produce analysis-ready tables	Aggregation, leakage-safe windows, target computation, feature engineering	Sale training set, rental training set, spatial market features

Table 1: Data-layer responsibilities.

4 Source Registry and Provenance

Every source must be registered before ingestion. Recommended metadata fields are:

```
source_id: dvf_geolocated
producer: data.gouv.fr / DGFiP derivative
source_url: https://www.data.gouv.fr/...
license: Licence Ouverte 2.0
snapshot_date: 2026-07-28
source_release: 2026-04
geography: France
coverage_start: 2021-01-01
```

```
coverage_end: 2025-12-31
retrieved_at_utc: 2026-08-07T16:00:00Z
sha256: "..."
schema_version: "1.0"
allowed_usage: "open-data reuse subject to licence"
notes: "Geolocated coordinates are parcel-based."
```


Mandatory provenance principle

A model artifact is not reproducible unless it can be traced to exact data snapshots, feature-code version, hyperparameters and training interval. Store these identifiers in MLflow or an equivalent registry.

5 Dataset 1 - DVF Transactions

5.1 Purpose

DVF (Demandes de valeurs foncières) provides the most important target data for the sale-price model. The official source is based on property mutations recorded by the French tax administration. The geolocated derivative on data.gouv.fr was refreshed for the April 2026 DVF release and was last updated on 28 July 2026. The official raw DVF publication states that the latest release was 7 April 2026 and that releases replace previously published files.[1]

5.2 What DVF contributes

Typical useful variables include:

- mutation date and nature;
- declared transaction value;
- address components;
- commune, postal code and department;
- cadastral parcel identifier;
- property type;
- built surface and principal rooms;
- land surface;
- longitude and latitude in geolocated derivatives.

The principal target is:

$$y_i^{sale} = \frac{V_i}{S_i}, \tag{6}$$

where V_i is the sale value and S_i the residential built surface.

[1] DVF official dataset: https://www.data.gouv.fr/datasets/demandes-de-valeurs-foncieres. Geolocated derivative: https://www.data.gouv.fr/datasets/demandes-de-valeurs-foncieres-geolocalisees. Accessed 7 August 2026.

5.3 Main data risks

DVF is not a simple one-row-per-home table

A mutation may contain an apartment, a cellar, a parking space, several lots or even multiple dwellings. Directly fitting a model to raw lines risks duplicating the price, misassigning surface and creating severe target noise.

Additional risks include:

- delayed recording: historical releases may receive late mutations;
- abnormal or non-market transactions;
- missing/zero built surface;
- multi-property mutations where a single declared value covers several dwellings;
- parcel-centroid coordinates rather than the exact apartment entrance;
- non-stable community identifiers across releases.

5.4 Preparation pipeline

Step 1: snapshot preservation. Save the untouched compressed source under a release-specific path:

```
data/bronze/dvf/release=2026-04/dvf_geolocated.csv.gz
```

Calculate SHA-256, file size and retrieval date.

Step 2: schema normalization. Cast dates, numeric prices, surface areas, room counts and coordinates explicitly. Treat textual codes such as postal codes and department codes as strings to preserve leading zeroes.

Step 3: geographic filtering. Keep the Ile-de-France departments:

$$D = \{75, 77, 78, 91, 92, 93, 94, 95\}. \tag{7}$$

Step 4: mutation profiling. Group raw lines by source mutation identifier and compute:

- number of apartment rows;
- number of house rows;
- number of dependency rows;
- number of parcels;
- total residential surface;
- unique property types.

Step 5: create modeling cohorts. The safest MVP cohort is a **simple single-dwelling sale**: exactly one apartment (or exactly one house in a separate model), positive price, valid surface, no conflicting residential unit.

Step 6: robust outlier filtering. Do not use one global price-per-square-meter threshold. Compute robust statistics within year × commune × property type:

$$z_i^{robust} = \frac{\log p_i - \text{median}(\log p)}{1.4826 \ \text{MAD}(\log p)}. \tag{8}$$

Flag rather than immediately delete extreme values; review their transaction structure.

Step 7: stable internal fingerprint. Generate an internal hash from normalized transaction attributes. Store both the original source identifier and internal identifier.

5.5 Derived features from DVF

Feature	Definition	Reason
`price_per_m2`	V/S	Primary normalized target
`transaction_year/month`	Extracted from date	Seasonality and time trend
`rooms_per_10m2`	rooms divided by surface	Compactness / layout proxy
`local_tx_count_12m`	Prior transactions within radius	Data density / liquidity
`local_median_ppm2_12m`	Prior local median	Strong local market anchor
`local_price_dispersion`	Robust spread of prior sales	Market heterogeneity
`distance_weighted_comparable_price`	Weighted by historical sales	Comparable-based valuation signal

5.6 Leakage-safe comparable feature

For transaction i at time t_i, only use transactions j satisfying $t_j < t_i$. A useful kernel is:

$$w_{ij} = \exp\left(-\frac{d_{ij}}{\tau_d}\right)\exp\left(-\frac{t_i - t_j}{\tau_t}\right), \tag{9}$$

with weighted local price:

$$\widehat{p}_i^{comp} = \frac{\sum_{j\in\mathcal{N}_i} w_{ij} p_j}{\sum_{j\in\mathcal{N}_i} w_{ij}}. \tag{10}$$

6 Dataset 2 - Base Adresse Nationale and Geocoding

6.1 Purpose

The **Base Adresse Nationale (BAN)** is the French address reference dataset. It provides standardized address entities and geospatial positions. In 2026, the former `api-adresse.data.gouv.fr` service is deprecated; the current public workflow is the Géoplateforme geocoding service. The official documentation indicates support for direct, reverse and bulk CSV geocoding, with an ordinary limit of 50 requests/IP/second; the BAN search engine is refreshed from BAN data twice per week.[2]

6.2 Role in the architecture

BAN is not a price source. It is an **identity and location bridge**. It converts inconsistent address strings into normalized address records and coordinates, enabling joins to parcels, buildings, DPE, transport and neighborhood layers.

6.3 Address normalization

Create both a source-preserving and normalized representation:

[2] BAN geocoding documentation: https://adresse.data.gouv.fr/outils/api-doc/adresse. BAN content: https://adresse.data.gouv.fr/contenu-de-la-ban. Accessed 7 August 2026.

```
raw_address       = "12 BIS AV. DE LA REPUBLIQUE 75011 PARIS"
normalized_number = "12"
normalized_suffix = "bis"
normalized_street = "avenue de la republique"
postal_code       = "75011"
commune_code      = "75111"
address_key       = hash(normalized fields)
```

Text normalization may include lowercasing, punctuation removal, whitespace collapse, standard abbreviation mapping and accent-insensitive comparison. Never overwrite the original string.

6.4 Geocoding quality

Store the geocoder's match score and returned object type. Recommended fields:

- `geocode_score`;
- `geocode_precision`;
- `geocode_method`;
- `returned_label`;
- `returned_citycode`;
- `latitude`, `longitude`;
- `geocoded_at`;
- `ban_snapshot_or_service_version`.

Do not silently accept low-score matches. A simple operational policy is:

$$\begin{aligned} S \geq 0.90 &: \text{high confidence,} \\ 0.75 \leq S < 0.90 &: \text{probable, retain quality flag,} \\ S < 0.75 &: \text{manual review or alternate matching method.} \end{aligned}$$

6.5 Bulk strategy

For millions of observations, deduplicate addresses first. Geocode each unique normalized address once, materialize an address reference table, then join back to all transaction rows.

6.6 Derived features

BAN enables:

- exact point geometry in WGS84;
- commune and district identifiers;
- spatial linkage to H3, IRIS, INSEE grids and parcels;
- address-level deduplication and temporal address aliases.

7 Dataset 3 - Cadastre and BDNB Buildings

7.1 Cadastre: spatial legal geometry

The open cadastral plan includes parcels, sections, building footprints and cartographic elements; owner/property files are not open. As of 7 August 2026, the portal offers a 1 June 2026 vintage in GeoJSON, Shapefile, GeoParquet and other formats, with download at several geographic granularities.[3]

7.2 BDNB: building-level consolidation

The **Base de Données Nationale des Bâtiments (BDNB)** is a building-level reference produced by CSTB through geospatial crossing of many public datasets. The May 2026 release describes roughly 32 million residential and tertiary buildings and is available in bulk formats and via APIs.[4]

7.3 Why both are useful

Dimension	Cadastre	BDNB
Primary role	Parcel/building geometry	Enriched building identity and attributes
Geometry	Parcels, sections, footprints	Building representation and linked geometry
Attributes	Limited open plan attributes	Construction period, uses, surfaces, energy-related and administrative variables depending on table/version
Join value	Stable parcel topology	Cross-dataset building identity
Best use	Spatial operations	Feature enrichment and building matching

Table 3: Complementary roles of Cadastre and BDNB.

7.4 Coordinate reference systems

Keep source geometries in their native CRS, but convert Ile-de-France calculations to Lambert-93 (EPSG:2154) for metric distances and areas. Never compute meter-based areas in longitude/latitude EPSG:4326.

7.5 Parcel and building matching

Recommended hierarchy:

1. exact source parcel identifier if available;
2. point-in-polygon against cadastral parcels;
3. parcel-to-building intersection;
4. nearest building centroid/footprint with distance threshold;
5. BDNB cross-reference, if available;
6. confidence flag when multiple candidate buildings exist.

7.6 Geometric features

For building b and parcel q:

$$Coverage_b = \frac{Area(Footprint_b)}{Area(Parcel_q)}, \tag{11}$$

$$Compactness_b = \frac{4\pi Area_b}{Perimeter_b^2}. \tag{12}$$

[3] Cadastre open data: https://cadastre.data.gouv.fr/; Etalab cadastral releases: https://cadastre.data.gouv.fr/datasets/cadastre-etalab. Accessed 7 August 2026.

[4] BDNB dataset: https://www.data.gouv.fr/datasets/base-de-donnees-nationale-des-batiments; BDNB portal: https://bdnb.io/. Accessed 7 August 2026.

Additional features:

- building footprint area;
- parcel area;
- number of buildings on parcel;
- distance from property point to building centroid;
- local building coverage within 250/500/1000 m;
- building orientation;
- adjacency/touching-building count;
- distance to street boundary where available;
- construction period and dwelling count from BDNB.

7.7 Quality flags

Store `building_match_method`, `building_match_score`, `building_candidate_count` and `geometry_valid`. Do not force a one-to-one building mapping when an address corresponds to a complex parcel.

8 Dataset 4 - DPE Energy Performance

8.1 Purpose and current coverage

The ADEME open dataset for existing housing contains DPEs established from 1 July 2021 onward. It is exposed through data.ademe.fr with tabular and API access.[5]

8.2 Useful variables

Depending on the current schema, useful families include:

- DPE label and GHG label;
- energy consumption and emissions;
- floor area;
- construction year/period;
- heating and domestic hot-water systems;
- insulation and envelope attributes;
- address/coordinates;
- assessment date and validity/status fields.

8.3 Important selection bias

DPE observations are not a random census of all dwellings. A DPE is frequently triggered by sale, rental, construction or other regulatory events. Therefore, missing DPE is not equivalent to poor energy performance. Add a `dpe_available` feature but do not interpret absence as a negative label.

8.4 Matching DPE to a property

The match should be probabilistic because exact shared identifiers may be absent. Candidate score:

$$S = 0.35S_{address} + 0.25S_{geo} + 0.20S_{surface} + 0.10S_{type} + 0.10S_{time}. \tag{13}$$

[5] ADEME DPE existing housing: https://data.ademe.fr/datasets/meg-83tjwtg8dyz4vv7h1dqe. Accessed 7 August 2026.

Prefer a direct building identifier relationship when BDNB provides it. Otherwise use normalized address, coordinate distance and surface similarity.

8.5 Temporal rule

For a valuation at time t, use only DPE information whose availability date is $\leq t$, unless the feature is explicitly treated as a static physical reconstruction. This distinction is essential for historical backtesting.

8.6 Feature engineering

Recommended features:

- ordinal DPE score $A = 7, \dots, G = 1$ (while retaining original category);
- energy-consumption intensity;
- GHG intensity;
- heating-energy category;
- construction-age bucket;
- `dpe_age_days` at valuation date;
- mismatch between reported transaction surface and DPE surface;
- building-level modal/median DPE when exact dwelling-level match is unavailable.

9 Dataset 5 - INSEE Demographic and Socioeconomic Data

9.1 Purpose

INSEE data describes the socioeconomic context around a property. The 2021 Filosofi gridded dataset published on 16 February 2026 includes 200 m cells and 34 variables on age structure, households, housing and income. Some cells contain imputed values and the imputation indicator must be retained.[6]

9.2 Levels of geography

Use multiple geographic levels because each captures different spatial behavior:

- commune/arrondissement;
- IRIS;
- 1 km grid;
- 200 m grid.

9.3 Potential features

At an aggregate area level:

- population and household density;
- income distribution indicators;
- owner/tenant composition;
- social-housing counts/shares where available;
- household size and age structure;
- housing composition.

[6] INSEE Filosofi 2021, 200 m grid: https://www.insee.fr/fr/statistiques/8735162?sommaire=8735243. Overview: https://www.insee.fr/fr/statistiques/8735243. Accessed 7 August 2026.

9.4 Statistical confidentiality and imputation

Small-area statistics may be suppressed or imputed to protect confidentiality. The 200 m Filosofi files explicitly identify imputed cells. Treat `is_imputed` as a quality variable and consider smoothing or fallback to a larger geographic level when microcell information is unreliable.

9.5 Temporal alignment

If the INSEE statistic describes year y, avoid applying it indiscriminately to earlier years. Recommended approach:

1. maintain an effective date interval for each demographic snapshot;
2. for transaction time t, select the latest dataset with reference period not after t where strict historical integrity is required;
3. alternatively, use a fixed census snapshot as a quasi-static structural feature but clearly label that experiment as non-causal.

9.6 Feature transforms

Use normalized rates instead of raw counts where possible:

$$OwnerRate_z = \frac{OwnerHouseholds_z}{Households_z}. \tag{14}$$

For skewed variables such as density or income, test log transforms and quantile bins. Avoid inferring individual resident attributes from area-level statistics.

10 Dataset 6 - Ile-de-France Mobilites

10.1 Purpose and format

Ile-de-France Mobilites (IDFM) provides the scheduled public-transport network in GTFS format. The official dataset covers planned service for the next 30 days across train, RER, metro, tram, bus and coach; the data are updated multiple times daily.[7]

10.2 Core GTFS tables

- `stops.txt`: stations and stops;
- `routes.txt`: lines/routes;
- `trips.txt`: scheduled trips;
- `stop_times.txt`: stop sequence and times;
- `calendar*.txt`: operating dates;
- `transfers.txt`: transfer relationships;
- `shapes.txt`: route geometries when available.

10.3 Feature maturity levels

Level 1 - Euclidean proximity. Distance to nearest metro, RER, train, tram and bus stop.

Level 2 - network density. Number of stations and distinct lines reachable within 500 m or 1 km.

[7] IDFM GTFS: https://prim.iledefrance-mobilites.fr/jeux-de-donnees/offre-horaires-tc-gtfs-idfm. Accessed 7 August 2026.

Level 3 - service intensity. Peak-hour departures, average headway and weekend/evening service.

Level 4 - door-to-destination travel time. Walking network + GTFS routing to business centers, universities and major transport hubs.

Level 5 - accessibility index.

$$Accessibility_i = \sum_j O_j e^{-\lambda T_{ij}}, \tag{15}$$

where O_j is opportunity mass (e.g., employment) and T_{ij} travel time.

10.4 Historical correctness

Current GTFS data must not be used to represent a 2021 transport network without qualification. Archive dated snapshots and maintain infrastructure opening/closure dates. For future infrastructure such as Grand Paris Express, separate current-accessibility features from planned-accessibility features.

11 Dataset 7 - OpenStreetMap and APUR Amenities

11.1 OpenStreetMap

OpenStreetMap (OSM) provides detailed volunteered geographic information: roads, buildings, shops, schools, parks, healthcare, leisure, pedestrian infrastructure and many other tagged features. OSM data are distributed under ODbL; public produced works require attribution, and database-derivative obligations should be reviewed for the intended product.[8]

11.2 APUR

The Atelier parisien d'urbanisme (APUR) publishes specialized open data for Paris and the Metropole du Grand Paris, including urban facilities and other local datasets.[9]

11.3 Why both

OSM provides broad and frequently updated coverage; APUR can provide curated Paris-specific classifications. Keep source provenance and prefer authoritative specialized sources when definitions matter.

11.4 POI taxonomy

Do not model thousands of raw OSM tags directly. Create a controlled taxonomy:

```
education:
  - school
  - university
  - nursery
retail_daily:
  - supermarket
  - convenience
  - bakery
health:
```

[8] OSM Foundation attribution and ODbL guidance: https://osmfoundation.org/wiki/Licence/Attribution_Guidelines. Accessed 7 August 2026.

[9] APUR open-data catalogue: https://www.apur.org/fr/donnees-disponibles-open-data; geographic catalogue: https://www.apur.org/fr/catalogue-donnees-geographiques. Accessed 7 August 2026.

```
  - hospital
  - clinic
  - pharmacy
green:
  - park
  - garden
  - forest
mobility:
  - bicycle_parking
  - bicycle_rental
  - pedestrian_zone
```

11.5 Spatial features

For each POI category k and radius r:

$$Count_{i,k,r} = \sum_j \mathbf{1}\{type_j = k,\ d(i,j) \le r\}, \tag{16}$$

$$Nearest_{i,k} = \min_{j:type_j=k} d(i,j). \tag{17}$$

Add area features for polygons, e.g. green area within 500 m, and network-distance features where pedestrian topology matters.

11.6 Non-linear proximity effects

Many amenities have non-monotonic effects. A train station can be desirable at 300 m but noisy at 10 m. Represent distance using bins, splines or tree-based models rather than forcing a linear coefficient.

11.7 Temporal bias

Today's OSM/APUR state may not describe a historical transaction date. Record source snapshot dates. For highly dynamic categories such as shops and restaurants, consider excluding them from strict historical backtests or using archived snapshots if available.

12 Dataset 8 - Rental Observations

12.1 Why a separate source is required

DVF provides sale mutations, not residential lease contracts. A rental model needs a distinct observed target. The Observatoire des Loyers de l'Agglomeration Parisienne (OLAP) publishes market and median rent information for Paris and surrounding areas, but aggregate observatory data are not equivalent to a complete apartment-level rental training corpus.[10]

12.2 Target definitions

Never mix different rental concepts without explicit labels:

- asking rent vs signed rent;
- rent excluding charges vs including charges;
- furnished vs unfurnished;

[10] OLAP portal: https://www.observatoire-des-loyers.fr/. The 2025 report states an average monthly rent of EUR 1,070 for 53 m^2 in the Paris agglomeration, or EUR 20.3/m^2: https://www.observatoire-des-loyers.fr/sites/default/files/news-files/Rapport%20Paris%202025.pdf. Accessed 7 August 2026.

- long-term residential lease vs short-term/seasonal rental;
- new lease vs sitting tenant.

Recommended normalized target:

$$y_i^{rent} = \frac{RentExclCharges_i}{Surface_i}. \tag{18}$$

12.3 Acquisition hierarchy

Preferred order:

1. signed leases from authorized partners/property managers;
2. licensed structured marketplace feeds;
3. agency or institutional landlord datasets;
4. public observatory/official aggregates for calibration;
5. only then, if legally permitted, public listing collection subject to terms, database rights and retention constraints.

12.4 Listing deduplication

The same dwelling may be reposted by several agents or after a rent change. Create a persistent listing/property entity using address, coordinates, surface, rooms, rent, text and image hashes where lawful.

A candidate duplicate score is:

$$S = 0.35S_{address} + 0.20S_{geo} + 0.15S_{surface} + 0.10S_{rent} + 0.10S_{text} + 0.10S_{image}. \tag{19}$$

12.5 High-value rental features

If listing history is available:

- first-seen and last-seen date;
- days online;
- number of price changes;
- initial vs final asking rent;
- duplicate-platform count;
- local active inventory;
- listing inflow/outflow per zone.

These are useful measures of rental market tightness but must be computed historically.

13 Dataset 9 - Macroeconomic Variables

13.1 Purpose

Macroeconomic variables capture the market regime at transaction time. For example, Banque de France reported an average rate of 3.21% for new housing loans excluding renegotiations in May 2026; these rates are updated monthly.[11]

[11] Banque de France, Crédits aux particuliers - 2026-05: https://www.banque-france.fr/fr/statistiques/credit/credits-aux-particuliers-2026-05. Accessed 7 August 2026.

The INSEE rent reference index (IRL) is published quarterly, while Eurostat publishes the House Price Index for residential properties purchased by households.[12]

13.2 Recommended variables

- mortgage interest rate;
- monthly mortgage credit production;
- consumer-price inflation;
- IRL level and year-on-year change;
- house-price index level/growth;
- unemployment rate;
- household disposable income where appropriate;
- construction-cost indices;
- building permits / housing starts;
- local transaction volume derived from DVF.

13.3 Point-in-time joins

The correct macro join is an **as-of join**. For a transaction on date t, select the latest value that was publicly available no later than t if the experiment is intended to simulate real-time knowledge.

This is different from joining by reference month after revisions. Store:

- reference period;
- publication date;
- revision/vintage if available;
- retrieved snapshot.

13.4 Nominal and real prices

For long-horizon modeling, maintain both nominal and inflation-adjusted targets:

$$RealPrice_t = NominalPrice_t \frac{CPI_{base}}{CPI_t}. \tag{20}$$

Do not train a short-horizon consumer-facing valuation model on real prices unless you explicitly convert the prediction back to nominal currency.

14 Cross-Dataset Entity Resolution

14.1 Canonical identity model

A robust project needs internal identifiers independent of source-specific keys:

```
address_id
parcel_id
building_id
property_id
transaction_id
rental_observation_id
source_record_id
```

[12] INSEE IRL: https://www.insee.fr/fr/statistiques/serie/001515333. Eurostat HPI: https://ec.europa.eu/eurostat/databrowser/view/prc_hpi_q/. Accessed 7 August 2026.

Relationships are many-to-many in raw sources. For example, one address can map to several buildings, one mutation to several parcels, and one building to many dwellings.

14.2 Matching hierarchy

1. exact source identifier join where semantics are trustworthy;
2. normalized deterministic join;
3. spatial containment/intersection;
4. probabilistic/fuzzy join;
5. manual-review queue for high-value ambiguous cases.

14.3 Match audit table

Every cross-source match should be auditable:

```sql
CREATE TABLE match_audit (
    left_source       TEXT,
    left_id           TEXT,
    right_source      TEXT,
    right_id          TEXT,
    match_method      TEXT,
    match_score       DOUBLE PRECISION,
    candidate_count   INTEGER,
    accepted          BOOLEAN,
    matched_at        TIMESTAMPTZ,
    pipeline_version  TEXT
);
```

15 Spatial Engineering

15.1 CRS policy

Use:

- EPSG:4326 for storage/interchange and web maps;
- EPSG:2154 (Lambert-93) for meter-based France distance/area calculations;
- H3 cells for hierarchical neighborhood indexing, while remembering H3 is an index rather than a cadastral boundary.

15.2 Neighborhood scales

No single radius is optimal. Create multi-scale features:

$$r \in \{100, 250, 500, 1000, 2000, 5000\} \text{ meters.} \tag{21}$$

This allows the model to learn street, neighborhood and wider submarket effects.

15.3 Boundary effects

A property 10 m from a commune boundary should not suddenly have completely different local features. Use continuous distance-based aggregations and H3/local-radius features alongside administrative categories.

16 Temporal Engineering and Leakage Prevention

16.1 The central rule

For a target observed at time t, no feature may use information generated after t unless the experiment explicitly represents a retrospective valuation.

16.2 Leakage examples

Feature	Wrong construction	Correct construction
Local price median	All transactions including future sales	Only transactions strictly before target date
DPE	Latest 2026 DPE for a 2022 backtest	DPE available on/before valuation date, or clearly labeled static reconstruction
Transport	2026 network for every historical year	Dated snapshots or opening-date-aware features
Macro	Final revised series joined by month	Point-in-time vintage or documented retrospective series
Rent inventory	Current listings	Listings active as of historical date

Table 4: Examples of temporal leakage and remedies.

16.3 Feature availability timestamp

Every Silver/Gold field with changing information should have or inherit an `available_from` timestamp. Historical joins then use:

$$Feature(i, t) = \max_{s \le t} \{FeatureSnapshot(i, s)\}. \tag{22}$$

17 Data Quality Framework

17.1 Quality dimensions

Measure at least:

- completeness;
- validity;
- uniqueness;
- consistency;
- freshness;
- spatial plausibility;
- temporal plausibility;
- match confidence.

17.2 Example validation rules

```
assert sale_price > 0
assert 5 <= surface_m2 <= 1000
assert rooms is None or 0 <= rooms <= 30
assert longitude is None or 1.0 <= longitude <= 3.8
```

```
assert latitude  is None or 48.0 <= latitude  <= 49.5
assert dpe_class is None or dpe_class in list("ABCDEFG")
assert transaction_date <= ingestion_date
```

Bounds are sanity filters, not final statistical outlier rules.

17.3 Quality dashboard

Track per source and release:

- row count change;
- missing-rate change by critical field;
- duplicate-rate change;
- match rates to BAN/BDNB/DPE;
- distribution drift in price, surface and coordinates;
- invalid geometries;
- freshness delay.

18 Feature Engineering Blueprint

18.1 Property-level features

- surface, rooms, property type;
- floor and amenities when available from rental/listing partners;
- DPE and heating;
- construction period;
- room density and surface bins.

18.2 Building-level features

- footprint, parcel area, coverage ratio;
- number of dwellings;
- building age/period;
- local building density;
- building-level energy aggregates.

18.3 Location features

- commune, arrondissement, IRIS and H3 embeddings/categories;
- distances to centers and major landmarks if justified;
- local comparable statistics at multiple radii;
- terrain/urban-form attributes where available.

18.4 Accessibility features

- nearest station by mode;
- walking/network travel time;
- line count and peak frequency;
- travel time to employment centers;
- accessibility indices.

18.5 Amenity features

- category counts and nearest distances;
- green-space area;
- retail/service diversity;
- negative externalities such as major road/rail proximity when data are available.

18.6 Market and time features

- local median sale/rent values using only prior observations;
- transaction/listing volume;
- price/rent momentum;
- mortgage rate and credit production;
- inflation and HPI/IRL trends;
- month/quarter seasonality.

19 Gold Table Design

19.1 Sale training table

Recommended columns:

```
transaction_id
valuation_date
sale_price
sale_price_per_m2
property_type
surface_m2
rooms
address_id
parcel_id
building_id
latitude
longitude
commune_code
iris_code
h3_r8
h3_r9
h3_r10
construction_period
dpe_class
energy_consumption
building_footprint_m2
parcel_area_m2
nearest_metro_m
nearest_rer_m
travel_time_chatelet_min
poi_retail_500m
poi_school_1000m
green_area_500m2
income_indicator_200m
insee_imputed
local_tx_count_12m_500m
local_median_ppm2_12m_500m
mortgage_rate
```

```
hpi_index
feature_snapshot_date
pipeline_version
```

19.2 Rental training table

Add:

```
rental_observation_id
rent_excl_charges
charges
rent_per_m2
furnished
listing_or_signed
first_seen
last_seen
days_online
asking_rent_initial
asking_rent_latest
reference_rent_zone
local_rental_inventory
```

20 Storage and Processing Strategy

20.1 Object storage

Use Parquet/GeoParquet for analytical data. Partition large time-series datasets by stable keys such as release date/year/department, avoiding very high-cardinality partitioning.

Example:

```
data/
  bronze/
    dvf/release=2026-04/
    dpe/snapshot=2026-08-01/
    idfm/snapshot=2026-08-07/
  silver/
    transactions/year=2025/department=75/
    buildings/department=75/
  gold/
    sale_features/feature_date=2026-08-07/
```

20.2 PostgreSQL/PostGIS

Use PostGIS for online/spatial lookup, canonical entities and indexes. Keep large immutable analytical scans in Parquet/DuckDB or a warehouse.

Recommended indexes:

```sql
CREATE INDEX idx_tx_geom ON transactions USING GIST (geom_l93);
CREATE INDEX idx_tx_date_commune ON transactions (transaction_date, commune_code);
CREATE INDEX idx_building_geom ON buildings USING GIST (geometry);
CREATE INDEX idx_address_norm ON addresses (normalized_address_hash);
```

21 Training Dataset Construction

21.1 Temporal split

Avoid random row splitting. Example:

- train: earliest years through 2023;
- validation: 2024;
- test: 2025 and later available observations.

Exact dates should reflect dataset coverage and business use.

21.2 Spatial split

Add a spatial stress test by holding out entire H3 cells, IRIS zones or communes. Also group records from the same building to avoid memorization leakage.

21.3 Baselines

Before advanced models, benchmark:

1. commune median price per m^2;
2. neighborhood/H3 median;
3. distance-weighted comparable estimator;
4. hedonic linear model;
5. CatBoost/LightGBM tabular baseline.

21.4 Metrics

Use:

$$MAE = \frac{1}{n}\sum_i |y_i - \hat{y}_i|, \tag{23}$$

$$MdAPE = \text{median}\left(\frac{|y_i - \hat{y}_i|}{y_i}\right), \tag{24}$$

$$RMSE_{log} = \sqrt{\frac{1}{n}\sum_i (\log y_i - \log \hat{y}_i)^2}. \tag{25}$$

Report metrics by department, commune, property type, price band, data-density band and time period.

22 Uncertainty and Confidence

Point estimates should be supplemented by uncertainty. Options include quantile regression and conformal prediction. A confidence score should incorporate:

- comparable count and recency;
- distance to nearest training observations;
- feature completeness;
- model ensemble agreement;
- out-of-distribution score;
- prediction interval width.

A confidence score is a product feature, not a substitute for a calibrated statistical interval.

23 MLOps Integration

23.1 Pipeline orchestration

Use Airflow or Dagster to orchestrate independent source pipelines. Each ingestion job should be idempotent and versioned.

23.2 Experiment tracking and registry

For every model version record:

```
model_name: sale_idf_catboost
model_version: 1.4.0
training_start: 2021-01-01
training_end: 2024-12-31
dvf_release: 2026-04
bdnb_release: 2026-02.a
dpe_snapshot: 2026-08-01
insee_snapshot: filosofi_2021_release_2026-02-16
feature_code_commit: abc123
validation_mdape: 0.091
coverage_90: 0.894
```

23.3 Monitoring

Monitor:

- source freshness and ingestion failures;
- schema drift;
- feature missingness and drift;
- prediction distribution drift;
- delayed ground-truth error when new DVF releases arrive;
- calibration of prediction intervals;
- API latency and failure rate.

24 Legal, Licensing and Governance Considerations

24.1 Open data is not licence-free data

Retain the exact licence and attribution requirements per source. Examples include Licence Ouverte for many French public datasets and ODbL for OpenStreetMap. APUR datasets may have dataset-specific licence terms. Rental-marketplace data can have contractual or database-right restrictions even if publicly viewable.

24.2 Personal and sensitive information

The valuation system should not attempt to identify buyers, sellers or tenants. Socioeconomic features should remain area-level. Avoid publishing raw records in a way that facilitates individual re-identification.

24.3 Feature governance

Maintain a feature catalogue with:

- feature definition;
- source(s);
- owner;
- transformation code;
- valid time / availability semantics;
- legal classification;
- training-serving availability;
- known limitations.

25 Implementation Roadmap

25.1 Phase 1 - Clean sale target

- ingest DVF snapshots;
- build mutation-level profiling;
- select simple apartment sales;
- establish robust target cleaning;
- create temporal/spatial baselines.

25.2 Phase 2 - Address and building identity

- BAN normalization/geocoding;
- Cadastre parcel joins;
- BDNB building enrichment;
- confidence-scored entity resolution.

25.3 Phase 3 - Physical and neighborhood enrichment

- DPE matching;
- INSEE 200 m/IRIS features;
- IDFM transport features;
- OSM/APUR amenity features.

25.4 Phase 4 - Rental system

- secure lawful property-level rent observations;
- normalize rent definitions;
- deduplicate listing histories;
- build separate rent target and calibration layers.

25.5 Phase 5 - Market and forecasting layer

- macroeconomic vintage store;
- local monthly market aggregates;
- 6/12/24/36-month forecasting models;
- scenario analysis.

26 Concrete Acceptance Criteria

A dataset release can be promoted from Silver to Gold only if:

1. source snapshot and checksum are recorded;
2. schema validation passes;
3. critical missingness is within documented thresholds;
4. duplicate/mutation logic has been validated;
5. coordinates pass spatial plausibility checks;
6. entity-match rates are measured and confidence stored;
7. all time-varying features respect point-in-time rules;
8. Gold features are reproducible from code and source versions;
9. training dataset leakage tests pass;
10. licence and permitted-use metadata are complete.

27 Recommended Repository Structure

```
real-estate-idf/
  configs/
    sources.yaml
    quality_rules.yaml
    feature_definitions.yaml
  data_contracts/
    dvf.yaml
    ban.yaml
    bdnb.yaml
    dpe.yaml
    rentals.yaml
  src/
    ingestion/
      dvf.py
      ban.py
      cadastre.py
      bdnb.py
      dpe.py
      insee.py
      idfm.py
      osm_apur.py
      rentals.py
      macro.py
    cleaning/
    entity_resolution/
    geospatial/
    features/
    quality/
    models/
    serving/
  pipelines/
    build_bronze.py
    build_silver.py
    build_gold_sale.py
    build_gold_rent.py
  tests/
  notebooks/
```

```
docs/
infrastructure/
pyproject.toml
docker-compose.yml
```

28 Final Integrated View

The information flow for one property can be summarized as:

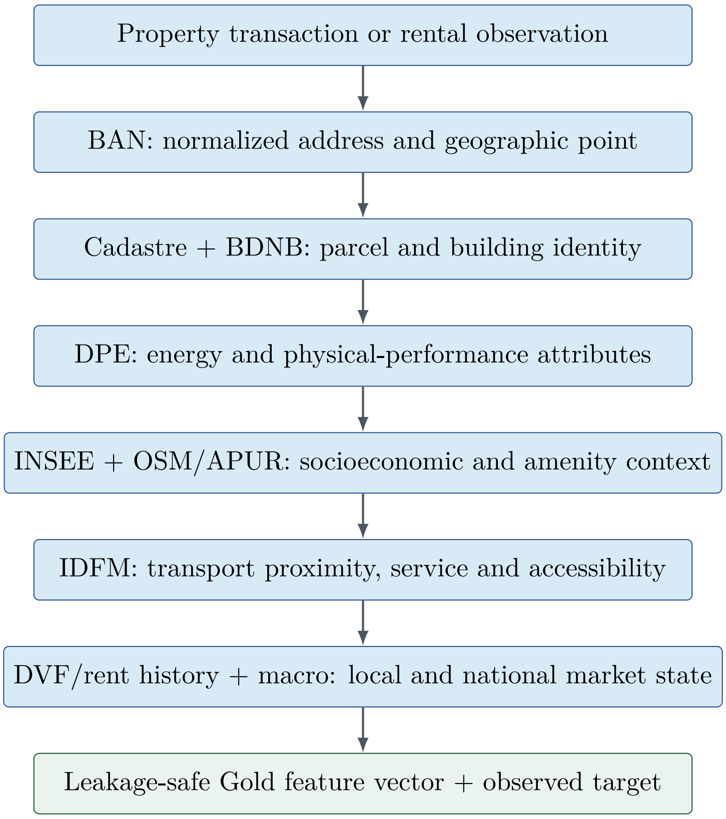

Figure 2: Integrated property reconstruction before model training.

The most important engineering insight is that the project is fundamentally a **data integration and point-in-time reconstruction problem**. Modeling quality depends on whether the system can correctly answer, for each historical property: what was sold or rented, where it was, which building it belonged to, what was known about the property and neighborhood at that time, and what the market environment looked like. A sophisticated model cannot compensate for incorrect transaction reconstruction, weak entity resolution, or temporal leakage.

29 Reference Sources

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[9] ADEME, DPE existing dwellings since July 2021: https://data.ademe.fr/datasets/meg-83tjwtg8dyz4vv7h1dqe
[10] INSEE, Filosofi 2021 gridded data: https://www.insee.fr/fr/statistiques/8735243
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[18] Eurostat, House Price Index: https://ec.europa.eu/eurostat/databrowser/view/prc_hpi_q/

Document Status

This deliverable is an implementation blueprint, not a substitute for the current schema documentation and licence text of each producer. API endpoints, quotas, field names and release schedules can change; ingestion code should therefore read current source metadata and preserve versioned snapshots.